

Supporting applications with network intelligence and B-ISDN

G L Bruce and J A Clark

The year 2000 is swiftly approaching and, with all the expectations it brings, many telcos are gearing up to provide telecommunications networks that will satisfy the majority of demands that new feature-rich applications will place upon them. To some people, supporting application requirements for a telecommunications network might mean adding more intelligence to the network. Others may crave to transmit data at faster speeds, communicate more freely on the move, or be able to mix and match the appropriate network infrastructure as they, or their applications, choose to do so. These diverse application requirements are shaping the strategies being pursued by telcos to deliver integrated telecommunications networks for the next millennium.

After looking back at some of the developments taking place within the broadband and network intelligence domains, this paper summarises some of the application and service requirements that must be addressed when building future telecommunications networks. From the plethora of technical proposals aimed at developing the necessary environment to fulfil these requirements, this paper explores the alternatives. Of course, for all of these alternative routes there will be many questions, technical or otherwise, that yet remain unanswered. Some of these issues will be discussed and various options will be considered to provoke thought or socialise possible solutions.

1. Introduction

There are many institutions working to develop what is essentially the global public network that we all presently use to transfer speech, data and images to one another. This global public network, the telephony network that we have today, has, in a large number of cases, reached its limitations and has been replaced, albeit on a non-global scale, by newer technologies capable of delivering information at faster speeds or in a wireless environment.

While these non-global networks are being successfully delivered, the utopia for many people working on these technologies must be to provide a global public network worthy of the 21st century. This network must accommodate fixed and mobile terminals in a consistent manner across a network, it must transport a multitude of media types, and it must provide the necessary logic to support and proliferate advanced services. The mandate for the people delivering this utopia is to piece together the innovation from a number of isolated network-specific solutions and to establish a totally integrated telecommunications network.

To understand the future global requirements this paper starts with the history surrounding the development of broadband ISDN (B-ISDN) and intelligent network (IN) recommendations produced by the Telecommunications

Sector of the International Telecommunications Union (ITU-T) (formally known as the CCITT). It continues with the service requirements from experts looking into areas such as broadband communications, mobile communications, intelligent networks and the Internet. It describes where we are now and what still needs to be done.

The introduction to section 4 describes some of the possible proposals being considered for the future global telecommunications network, and delivers the rationale for proposing that the integration of B-ISDN with network intelligence will provide the best way forward. The rest of the section then describes two parallel routes that development processes are taking towards an integrated telecommunications network. Both of these routes are based on B-ISDN and IN technologies; however, while one route is based upon a nearer term adaptation of the equipment and technology we have at present, the other route is based on a longer term evolution of today's equipment and technology. Perhaps, not surprisingly, the nearer term route is somewhat driven by the equipment manufacturers while the longer term solution is largely being promoted by telcos within the ITU-T.

The paper concludes by speculating on how close telcos will be to meeting user requirements for the 21st century. It predicts that integrated telecommunications networks will

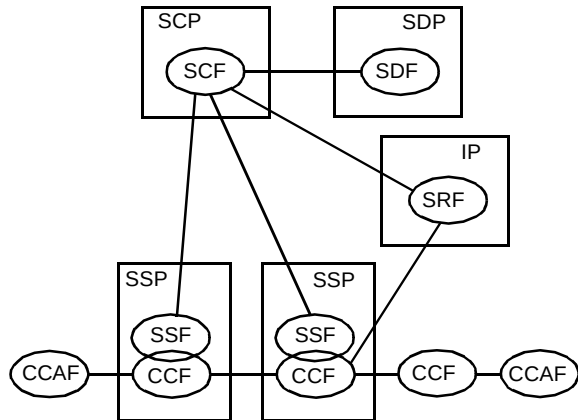
not arrive until five years into the next century and suggests what the likely stop-gap will be.

2. The development of B-ISDN and IN

2.1 The intelligent network

The IN can be considered as the precursor to multi-service information networks as it was the first time the network had been used for services other than connections between two locations. It provides advanced services which cannot be easily implemented in the switch or are best offered from a central point in the network. To achieve this, the digital switch is enhanced so that it becomes an IN service switching point (SSP), thus enabling it to notify the service control point (SCP) when processing of the call/connection requires an IN interaction.

The IN functional architecture is illustrated in Fig 1. The existing software which controls the switch and communicates with the terminal is the call control function (CCF). The service switching function (SSF) is added to provide the interface to enable call processing to interact with the IN control platform. The flexibility of the IN arises from the SCP which comprises a service control function (SCF) and a service data function (SDF). The SCF runs services via service logic programs (SLPs). The SDF provides service information like translation tables and user profile data. The specialised resource function (SRF) can be temporarily connected to the caller to play announcements and collect digits from them.



SCP service control point
 SCF service control function
 SDF service data function
 SSP service switching point
 SSF service switching function
 CCF call control function
 CCAF call control agent function
 SRF specialised resource function
 IP intelligent peripheral

Fig 1 IN CS-1 functional architecture.

The ITU-T are developing and releasing IN specifications in stages known as capability sets.

IN CS-1 [1] was first released in 1992 and contains the capabilities to re-route calls, modify charging or billing

records, and obtain extra information from the caller via the SRF. Service-independent building blocks (SIBs) are used to construct IN services in SLPs. Ideally a service built from SIBs would be compiled and run on all SCP platforms. In practice, the different SCP manufacturers have produced proprietary service creation systems (SCSs) which only adopt SIB principles. The SLPs communicate with the underlying SSPs via INAP (intelligent network application part) operations. As it is easier to compare INAP operations with the requirements for B-ISDN, this paper will concentrate on limitations in this area as opposed to more generic SIBs.

A list of IN CS-1 INAP operations is given in Table 1. This should enable the capabilities of IN CS-1, and the services that can be built up, to be envisaged.

Table 1 IN CS-1 SSP, SCP and SRF operations.

INAP operation	Action
Activate service filtering/response	Sets up event monitoring, e.g. televotes
Activity test/response	Test for continued SSF/SCF relationship
Apply charging/report	Requests charging data from SSF
Call gap	Restricts requests to SCF for overload control
Call information request/cancel/report	Generates call report at end of call
Collect information	Collect more digits
Connect	Requests the SSP to connect the call up to the specified destination
Connect to resource	Connects call to SRF, SSF relays all commands to and from SRF
Disconnect forward connection	Disconnects call from SRF
Establish temporary connection	Connect call to assisting SSF, intelligent peripheral (IP) or separate SRF
Event report BCSM (basic call state model)	Report of a dynamic event
Furnish charging information	Provide charging information for SSF call records
Initial detection point (DP)	Initial service request for SCP
Initiate call attempt	New call initiated by IN
Play announcement	Plays an announcement on SRF
Prompt and collect	Plays an announcement and collects digits on SRF
Release call	Clears down call
Request notification charging event/event notification charging	SCF requests notification of charging events
Request report BCSM event	Set up monitoring for dynamic event
Send charging information	Allows SCF to set charging rate (e.g. free)

IN CS-2 [2] contents have recently been completed. To illustrate the new capabilities, a list of new IN CS-2 INAP operations are given in Table 2.

Table 2 New IN CS-2 SSP, SCP and SRF operations.

INAP operation	Action
Activate/deactivate trigger data/confirmation	Allows SCF to set up static events (triggers)
Create call segment association (CSA)	Create a new CSA, does not contain any call segment (CS) initially
Disconnect leg	Disconnects a specific leg, other legs remain in call
Event report facility	Reports that an appropriate DSS1 facility IE has been received
Hold call in network	Used for IN call queuing systems
Join	Joins one call into an existing call
Merge call segment	Connect call segments together
Move call segment	A split operation followed by a join on a call segment
Move leg	Moves legs between call segments
Prompt and receive message	Request SRF to play announcement and store a message
Reconnect	Re-establishes communication between a controlling leg and held party
Request/report UTSI	Request/report user to service information (UTSI)
Request/report BCUSM event (request/report basic call unrelated state model event)	Requests SSF to monitor for certain call unrelated events from the user
Script run/event/close	Commands to control user scripts on SRF
Send STUI	Send service to user information (STUI)
Split leg	Separates one party from its call segment and places it in its own associated CS

The most significant change is the addition of leg control known in IN CS-2 as call party handling (CPH). It enables individual parties to be joined into existing calls to create multi-party calls. Individual or multiple parties (call segments) can be split up and reconnected with other calls. The correct terminology and a full definition can be found in the ITU Q.1224 specification [2].

Call party handling and B-ISDN
It should be noted that these CPH enhancements go some way to addressing the multiparty/multi-connection requirements in the B-ISDN. Making them fully consistent with B-ISDN and backwards compatible will be the key task for an integrated IN/B-ISDN network (see section 2.2, Table 4).

Other significant additions to IN CS-2 include call unrelated signalling which can be used to enable terminal-to-IN communication without having to set up a connection to an SRF. Also there is a new SCF-to-SCF interface to enable service control information to be exchanged between

network operators. This avoids all calls/connections being tromboned back into the caller's home network. The SRF has also been improved to enable scripts to be run, and text information to be sent and received from ISDN phones, and the ability to store messages will enable network-based answering services to be built into the IN.

In relation to broadband these new features provide very useful capabilities which, with a few enhancements, could be exploited to enable a service provider to set up multiple connections per call as required for videoconferencing, broadcast TV or many other services which are not just single end-to-end connections.

2.2 The broadband network (B-ISDN)

During the 1980s, telcos were investigating how to develop their networks beyond the plain old telephony service (POTS) that we know today [3]. In 1988, asynchronous transfer mode (ATM) heralded a revolution that made the leap from narrowband services to broadband services a possibility. At around the same time the ITU-T drafted a framework to develop B-ISDN. This framework consisted of three phases, called releases. As B-ISDN is conceptually providing no more than higher speed N-ISDN services, B-ISDN release 1 was targeted simply to adapt the N-ISDN services, such as telephony and constant bit rate services, to operate on ATM technology. Table 3 gives the high-level requirements for B-ISDN release 1. The immediately obvious similarity of B-ISDN release 1 with N-ISDN is that the call and network connection are tied together when a 'call/connection' is established or cleared down. The other comparison with N-ISDN is that only single-connection, point-to-point 'call/connections' can be configured. As mentioned earlier, B-ISDN release 1 does offer larger bandwidth 'call/connections'; however, it also supports the additional provision for variable bit rate 'call/connections' as an alternative to the narrowband-like constant bit rate 'call/connections'.

Table 3 B-ISDN release 1 signalling requirements.

Signalling requirements	Connection type
Simultaneous call and network connection establishment	Point-to-point, single-connection
Release of a call and associated network connection	

The signalling recommendations for B-ISDN release 1 were published early in 1995 and network equipment supporting these recommendations is now available. Support for ITU-T B-ISDN terminal equipment, however, is somewhat lagging, the majority of terminal vendors preferring to build their equipment on ATM Forum standards.

Releases 2 and beyond of B-ISDN were then identified to cover new and enhanced services, such as broadcast services, video-on-demand (VoD) and videotelephony. To support some of these extra services, the main innovation for B-ISDN release 2 is the functional separation of call control and bearer control [4], to support the new connection configurations required by these new services. Table 4 depicts the high-level signalling requirements for B-ISDN release 2. It has been greatly simplified to give a taste of the possible richness of B-ISDN release 2 capabilities [5]. New capabilities include point-to-multipoint network connections, the ability to support multiple network connections within a call, plus the ability to manipulate the calls independently of the network connection(s) and vice versa. Also worthy of note is that the capabilities listed in Table 4 can be invoked by any party within the call, even if they are not associated with a network connection.

Protocol pitfalls

Although the point-to-multipoint protocol Q.2971 [6] was developed in the B-ISDN release 2 time-frame, it is the opinion of the authors that in the haste to provide the point-to-multipoint capability it failed to address the call and bearer control separation issue. As a result, additional network connections cannot be added after a Q.2971 call/connection is established because it does not support both call reference and network connection values. The 'call ref' tag used in Q.2971 is a carry over from the time when a call always contained a single connection and as such the 'call ref' could be construed as referring to a connection with an implied reference to the call. In a separated call and bearer environment it is likely that the 'call ref' will be used as a network connection reference while a new tag will be generated for a call reference.

Q.2931 [7] is an even more restricted version of Q.2971 [6] and there are current moves to revise this protocol to accommodate call and bearer control separation concepts.

While the signalling requirements for B-ISDN release 2 were completed in December 1993, the sheer complexity of providing these additional capabilities led the ITU-T into delivering the signalling recommendations for this release in two main steps. The majority of the signalling recommendations for the first step of B-ISDN release 2 have been approved and approval for the remaining recommendations will follow shortly. Network or terminal equipment supporting some of these recommendations has yet to materialise beyond trial versions. The signalling recommendations for the second step of B-ISDN release 2 are still being developed, and it is this step that contains the activity to support separation of call control and bearer control. It is expected that approved versions of these recommendations will start to become available in May 1998.

Table 4 B-ISDN release 2 signalling requirements.

Signalling requirements	Connection type
Simultaneous call and network connection establishment	Point-to-point, multiconnection
Call establishment without network connections	
Addition of one or more new network connections to an existing call	
Addition of one or more new parties to an existing call	
Attach one or more existing parties to an existing network connection	
Detach one party from an existing network connection	or
Release a network connection from an existing call	Point-to-multipoint multiconnection
Release a party from an existing call	
Release a call and associated network connections	
Renegotiation and modification of an active network connection	
Look-ahead at the destination party's availability and terminal characteristics	

The primary objective for B-ISDN release 3 was labelled as the integration of B-ISDN with IN, mobile communications and control systems for operating and managing the network. The first phase of the signalling requirements for B-ISDN release 3 are expected to become stable by October 1998, with signalling recommendations for this phase starting to become available in February 2000. The second, and possibly final, phase of the signalling requirements for B-ISDN release 3 are expected to become stable by September 1999, with signalling recommendations for this phase starting to become available in June 2001. The high-level B-ISDN release 3 signalling requirements are given in Table 5 in section 4.3.4.

3. Application requirements for future telecommunications networks

There are many changes taking place in the type and volume of applications being run by networked terminals. These changes have partly been driven by the high-speed, low-cost, wide area networks to which the terminals are connected, and partly by the advances in terminal technology. Requirements for new applications have mainly originated from the IT world; however, the main telecommunications players are not far behind.

The next few sub-sections describe some of the main application, or service, requirements coming from experts working in the field of intelligent networks, broadband ISDN, the Internet, IT and mobile networks. Each set of requirements presents its own obstacles that must be crossed. What will be discussed is which of the requirements have already been achieved and which are currently being addressed. Section 4 draws all of these requirements together and examines what needs to be done to fulfil these requirements.

3.1 *IN services*

Services that can be constructed on an IN platform include routing to non-geographic numbers, such as freefone and premium rate numbers, virtual private networks, call screening, closed user groups, call distribution according to load, time of day, authentication for charge cards or account calls, and generation of advanced billing records. Most of these services will still be applicable in a broadband network and there are abundant opportunities for many more services.

The new broadband services are likely to provide new requirements from service providers. These will include the ability to set up, release and move connections to their customer. For example, offering an application programming interface (API) to the service control for service providers could enable VoD servers to be distributed around the network while still remaining under the control of the service providers.

The evolution to broadband will also require a number of enhancements to the SRF. It may now be acting as a service gateway to which the user initially connects in order to choose the desired service. It will therefore need to have the ability to play MPEG data streams, retrieve responses and download software to users' set-top boxes or PCs.

3.2 *B-ISDN services*

The main requirements coming into the B-ISDN domain have come from the Digital Audio-Visual Council (DAVIC) [8]. Requirements from DAVIC have been focusing on the emergence of new multimedia services such as VoD, broadcast and staggercast TV with return channels, videoconferencing and videotelephony. While VoD, broadcast and staggercast have been specified within DAVIC, videoconferencing and videotelephony have as yet to be specified.

Within DAVIC, services such as VoD have led to the development of a session control plane above the call/connection control plane as a means to co-ordinate disparate call/connections between the parties involved in long-lasting sessions.

Within the ITU-T, the VoD-type services have led to the development of session control and resource control to provide a means to co-ordinate calls and connections not only within a particular network implementation but also across many disparate network implementations. Within the ITU-T, session control is network implementation independent. This network independence is provided by resource control. It will provide functions similar to those provided by WinsockTM and H.245 [9] in that it will provide a single API to session control while adapting to different network implementations (i.e. ITU-T B-ISDN, ATM-F B-ISDN, POTS, Internet) and capabilities (i.e. Q.2931, Q.2971) over which it is running. Extra flexibility being built into session control will also allow end applications

involved within a session to negotiate for access network and software resources when these resources are not readily available.

3.3 *IP/Internet applications*

The massive growth of the Internet has brought about a wealth of new requirements placed on the 'net' from more sophisticated applications. Gone are the days of simple UDP (user datagram protocol) and TCP (transmission control protocol) support for applications such as file transfer programs and browsers. Newer applications such as NetMeetingTM, which supports videoconferencing on a non-guaranteed quality of service (QoS) using the ITU-T Recommendation H.323 [10], would provide a better service if transported using a guaranteed QoS. Today, services such as the Internet phone are starting to demand a guaranteed QoS, and it is for this reason that the newer capabilities such as RSVP (resource reservation protocol) [11] are being developed for the Internet.

As with most telecommunications systems, the Internet is both a threat and an opportunity for the existing intelligent network. There are no formal IP interfaces specified as part of the IN architecture although some types of SCP do offer this capability. It could be used to provide customers with access to their IN data via the Internet or for dial-in PC connections. Other possibilities include CTI-type applications where extra information about the caller is presented on the screen of the person answering the call.

Another area is the IETF PSTN/Internet Interfaces (PINT) Working Group which is addressing connection arrangements through which Internet applications can enhance PSTN services. An example of such a service is a Web-based Yellow Pages service with the ability to initiate PSTN calls between customers and suppliers. This will require new APIs between the PSTN and the Internet.

Key new requirement

It is the authors' opinion that some form of Internet/IP interface to the SCF is required in future IN specifications. This would significantly improve the capabilities and flexibility of the IN and the services that can be operated from an SCP.

3.4 *Mobile services*

A mobile terminal's requirements should be the same as those pertaining to a fixed terminal such as basic connectivity and IN support for enhanced services. However, there are a few notable exceptions, the main one being that the mobile terminal should be allowed to roam between base stations or even exchanges [12]. Although roaming is currently achievable using GSM, a more resource-efficient method is being specified within the International Mobile Telecommunications (IMT) 2000 initiative. The other main difference is that the total possible bandwidth delivered to a mobile terminal is going to be less

than that delivered to a fixed terminal. While this apparently reduced bandwidth to mobile terminals is likely to imply a subset of services compared with those provided to fixed counterparts, it is this 'bandwidth challenge' that is inspiring services such as SMS, and no doubt others that are unique to the mobile domain will follow.

4. Alternative approaches to building an integrated telecommunications network

4.1 Proposed solutions for future telecommunications networks

Section 3 examined the pressure placed upon any future public network by a number of dissimilar but potentially overlapping user requirements. How to realise a future telecommunications network that would fulfil all of these requirements would be impossible to predict. At one extreme the perfect solution might be to build a distributed environment in which all terminals and applications can exist, while at the other extreme the most pragmatic solution might be to enhance the present terminals and applications that run on the Internet. Either one of these proposed solutions could, with some extra effort, quite easily be the dominant future global telecommunications network. The immaturity of object buses in a global telecommunications environment and the issues of quality of service, security and charging surrounding the Internet possibly makes B-ISDN based solutions the most likely of contenders for future public telecommunications networks. B-ISDN is a mature and proven technology that, among other things, provides quality of service, security and charging. However, B-ISDN is not the complete panacea for all user requirements as there is still much work to be done.

This section poses two alternative approaches to building an intelligent broadband network for the future. The first approach looks at a near-term realisation by adapting and enhancing the equipment that is presently available to solve user requirements as and when they emerge. The second approach is a longer term realisation of the unified telecommunications network that has been based upon the initial requirements for B-ISDN release 3 and enhanced along the way.

4.2 Nearer term realisation to building an integrated telecommunications network

Although a lot of the concepts presented here are a long way from being realised in hardware and software, there are a number of services which can be built from systems available now or in the near future. This section covers some of these solutions to illustrate how the fast pace of telecommunications development is stimulating new services to be constructed from current technology.

4.2.1 DAVIC IP-browsing user interface

This is a good example of using existing technology to solve the current limitations of user interfaces to the network. The terminal now contains a Web browser (i.e. Netscape or Internet Explorer) and this talks to a call control function which is basically a WWW server. The advantage this gives is that the user can be presented with an easy-to-use graphical interface that can be customised for the particular services to which a user subscribes. Figure 2 illustrates this solution.

The broadband access network connects the set-top unit (STU) to the Internet service access point (ISAP) via a variety of technological possibilities not covered here. The HTTP request is for the reserved address of the HTTP server, which is providing the equivalent of the network intelligence layer. The user will be presented with a range of service options and selects a video which is provided by the application server shown in Fig 2.

The HTTP server downloads a DSM-CC user-to-user [8] client Java applet which is needed for accessing the controls on the application server. This applet will convert commands from the PC/TV remote control (play, stop, FF, etc) into DSM-CC U-U messages. The connection for the MPEG-2 stream flow is made via DSM-CC user-to-network (U-N) commands from the STU initiated by the Java applet. Alternatives are feasible, such as setting up this connection from the HTTP server which would significantly simplify protocols supported by the STU at the expense of complexity of the switch-to-server interface.

4.2.2 Alternatives to INAP

Instead of waiting for the INAP specifications to be enhanced to support all the capabilities required and for ATM switches to be upgraded to support INAP, some alternatives are being explored. These include using interfaces to the switch management system or using a proxy signalling interface to set up the connections across the switch.

Proxy signalling was originally developed to allow set-top boxes with minimal signalling capability to set up calls on the network. They would communicate with a proxy signalling agent in the local network, which would support the full requirements of a broadband edge switch and set up the call on its behalf.

At the moment, providing a UNI on ATM switches to enable users to set up their own ATM connections (switched virtual circuits) is a higher priority than adding IN interfaces. As proxy signalling is only an enhancement to UNI signalling, this interface is currently more realisable than an ATM IN switch.

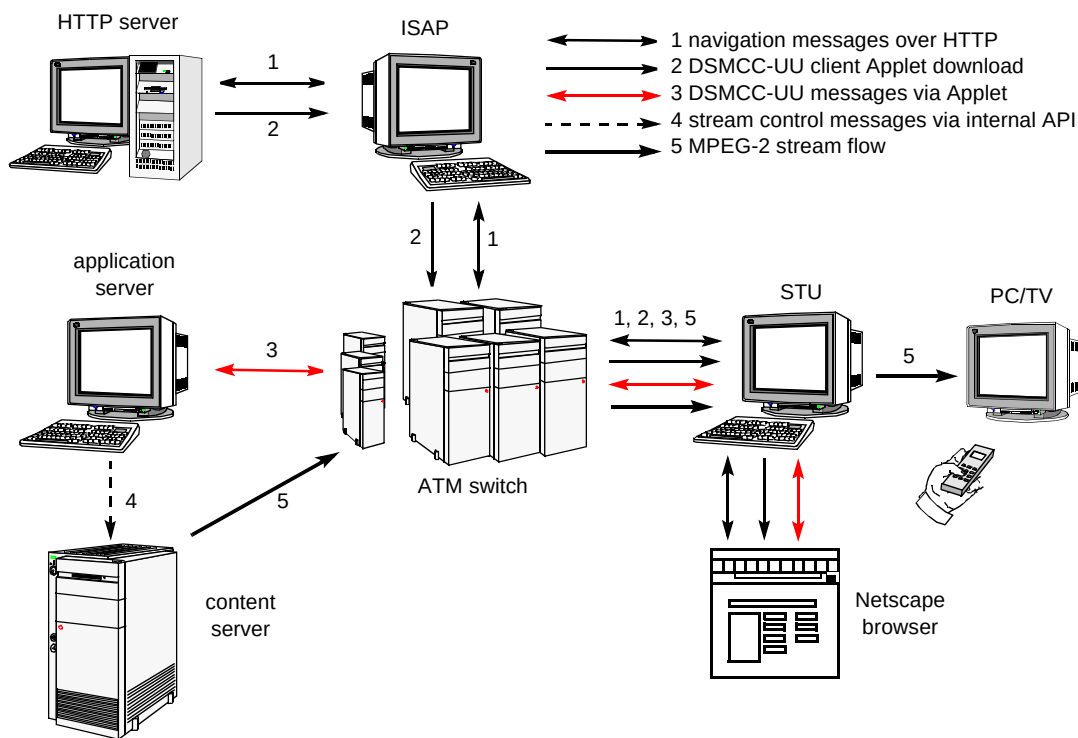


Fig 2 IP browsing user interface.

The limitations of this approach relate to the user-to-IN interface which is not clear because of the lack of IN basic call state models to represent the call set-up states. This may be overcome by terminating the user signalling in the IN SCP instead of the CCF, although the browsing interface is particularly appropriate for this purpose.

4.3 Longer term realisation to building an integrated telecommunications network

Within the ITU-T the goal is for a truly unified network that is rich in network capabilities for all end users. This vision of a 'unified telecommunications network', although based on B-ISDN release 3 requirements, has also been influenced by the Internet, DAVIC, TINA-C and multimedia groups.

To progress the 'unified telecommunications network' the ITU-T is currently using a method to develop B-ISDN and IN signalling requirements and hence capabilities. It is expected that future mobility and multimedia signalling requirements will be developed using the same method. The

method used is derived from I.130 [13] and object-oriented techniques, especially the Open Distributed Process Reference Model (ODP-RM) [14], to provide a co-ordinated framework within which to develop the standards. Figure 3 shows an overview of the framework.

Having acquired service requirements and generated the service definitions, a business model is generated to provide a common platform of understanding for the experts from different disciplines involved in pulling together the unified network. Precise definition of the data and behaviour of the system at a fine granularity object level is carried out. This shows information that must be passed around the system when each of the capabilities of the system is invoked. Using the object model, the objects in the telecommunications 'system' are partitioned into coarser granular objects which make up the functional model. The necessary information exchanges around this system are expressed as information flows and functional entity actions which are produced to verify the analysis and the functional model. These show how entities co-operate and what information is passed between co-operating entities in order to realise a

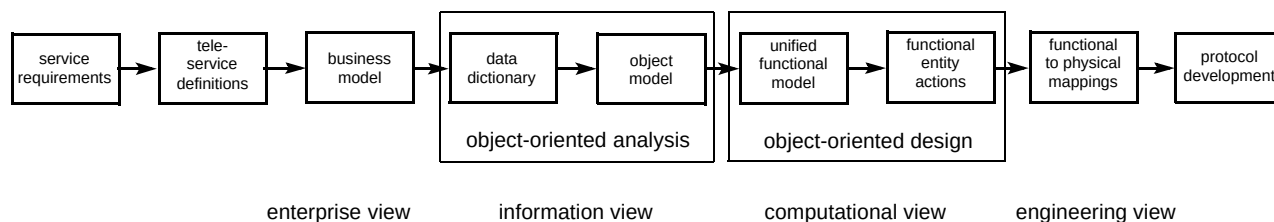


Fig 3 Framework to develop unified network standards

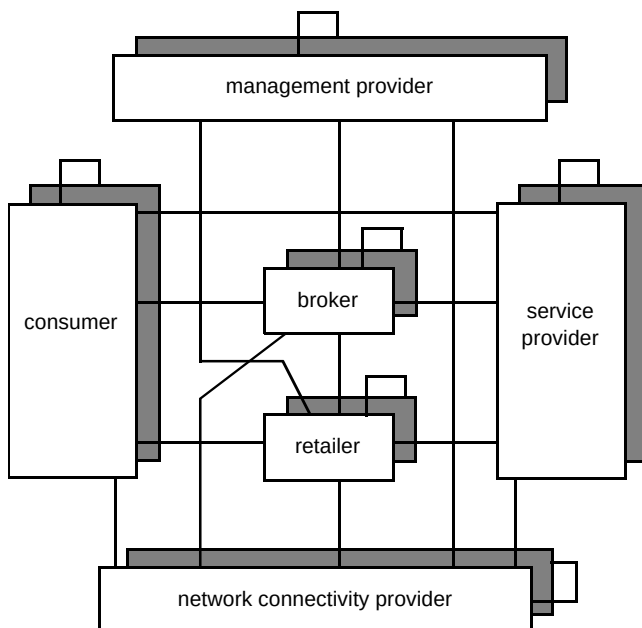
capability. At this point, this information could be passed on in order to generate the necessary protocols; however, as protocols only need to be standardised at physical interfaces, functional entities are mapped on to physical entities to identify physical interfaces and to reduce the number of protocols needing to be standardised.

The following sections show in detail some of the more interesting processes and results of this framework, namely the business model, the functional model, functional entity actions and the new capabilities.

4.3.1 The business model

The business model captures business requirements to structure and orient the system of interest. The model is derived from the open distributed processing (ODP) enterprise viewpoint [14], to partition the total business enterprise into business domains. These business domains represent, for example, the users, owners and providers of information processed by the system.

The proposed business model is shown in Fig 4 and is composed of six types of business domain, connected by inter-domain logical reference points. These logical reference points identify the communications channels available to exchange information between these business domains.



The model illustrates the following business domains:

- consumer — enrolls for and consumes information (a consumer may be an individual, a household with a small number of end users, or a business with multiple end users),
- broker — provides location information that the business domains use to locate other entities for the purpose of establishing communications,

- retailer — provides the optional initial contact point for the consumer to arrange for, and the contact point for the service provider to offer, services which employ communication services as the delivery mechanism,
- service provider — provides any of a variety of services which employ communications services as the delivery mechanism,
- network connectivity provider — provides the communications services which transport information that may either be control plane information or user plane information,
- management provider — manages, in whole or part, a number of business domains and provides the necessary administrative capabilities.

There is nothing in the model that precludes any combination of these six domains within a single physical entity. For example, an IN SCP which may typically be associated within the broker domain, may contain functionality specific to the retailer domain.

4.3.2 The unified functional model

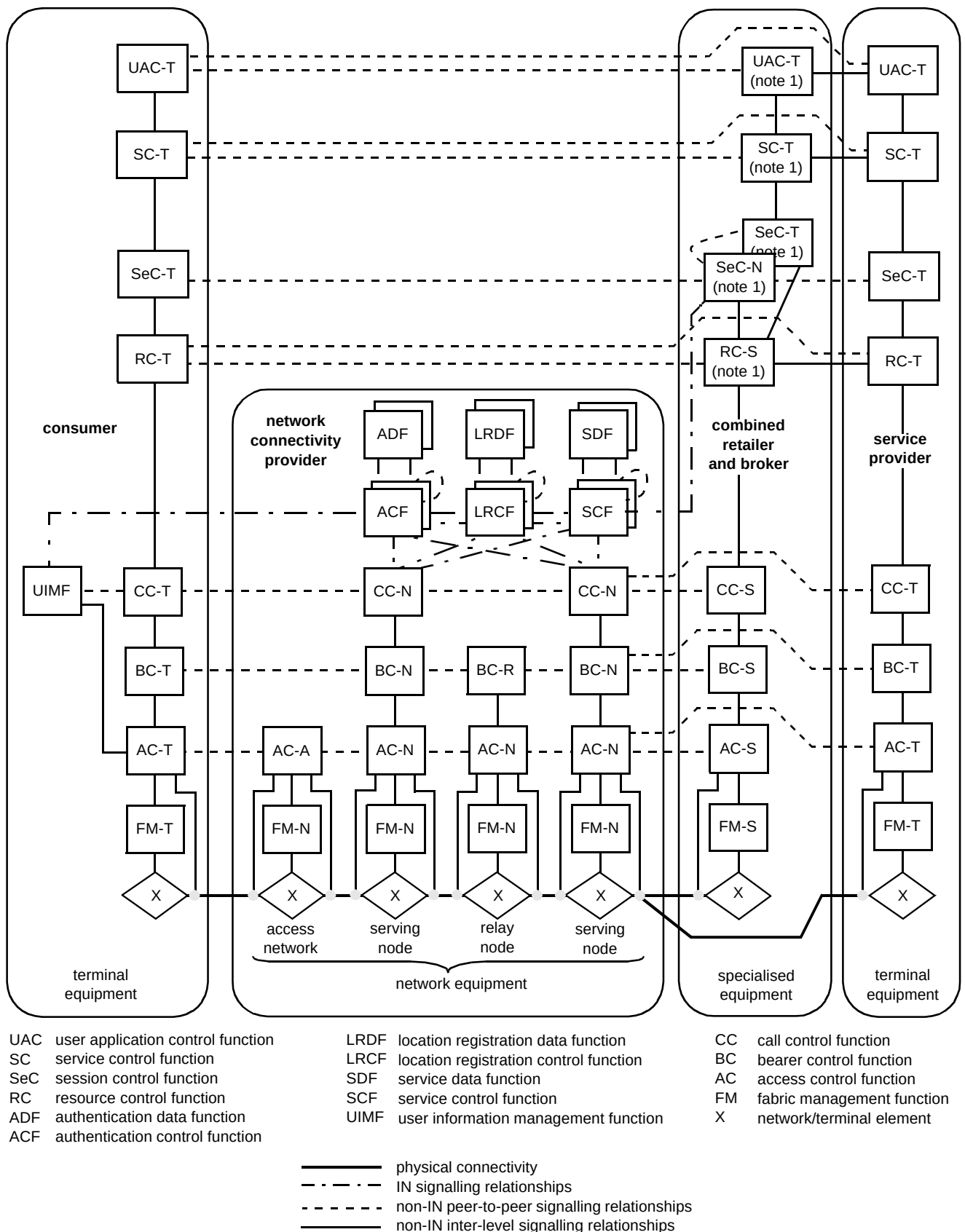
The unified functional model (UFM) partitions the system into functional entities (FEs) performing individual functions that will be conducive to flexibility in support of new services and in deployment of service processing in the network. The model combines and integrates the aspects of the IN distributed functional plane (DFP) architecture [2], the IMT 2000 functional network architecture (FNA) and the B-ISDN architecture [15].

The proposed model is shown in Fig 5 and is expressed as FEs which communicate via FE relationships. The FEs are independent of physical deployment and implementation in any of the business domains, providing the flexibility for multiple physical network configurations.

The model has been arranged to show one possibility of how the functional entities map on to the business model. Although complex in its functionality, the salient components of the unified functional model, however, are described below.

The UFM is best described by dividing it into the vertical functional groupings given below:

- network/terminal element (X) — although not strictly an FE, the network/terminal element represents an aggregation of network or terminal elements under control of a single functional entity,
- fabric management (FM) — manages the resources contained within a particular network or terminal element,



Note 1 These functions could be emulated by the SCF; in such cases, the relationships towards these functions would lead to the SCF

Fig 5 The unified functional model overlaid upon the business model.

- access control (AC) — provides access channel control to establish, for example, a signalling channel on a mobile air interface,
- bearer control (BC) — supports the control of network resources on a network link by network link scale to provide end-to-end carriage of information,
- call control (CC) — provides the control of functions on a network-wide scale (CC is not technology dependent and can be used over a number of different concatenated transport technologies to co-ordinate a call),
- resource control (RC) — uses the appropriate network infrastructure to provide the capabilities required by session control, with this network infrastructure possibly being an ATM network, Internet, wireless, etc (note this RC is not the same as resource control defined in MAGIC [16]),
- session control (SeC) — supports the control of communications sessions between user applications, including end-user (customer), network (broker) and third-party (retailer) based applications, and, through an API, also supports a single instance of a telecommunications application,
- service control (SC) — provides end-to-end brokerage-type functions between terminals, an example of which could be for an advertisement of the identifier for a broadcast or conferencing service required by third parties prior to originating a leaf-initiated join to the call (the need for this functionality is currently undergoing investigation),
- user application control (UAC) — provides the control of the application itself, e.g. pause, rewind, bookmarking and content selection for a VoD application.

Additional support functions contained in the UFM are:

- user information management function (UIMF) — provides the end user's service profile to allow for service mobility,
- access control/data functions (ACF/ADF) and location registration control/data functions (LRCF/LRDF) — perform processing and provide access to data that is specialised for particular non-call-related (call-unrelated) mobility applications — it is likely that the ACF and LRCF will be merged into a coarser-grained function possibly to be called a call-unrelated service control function (CUSCF), and the same applies to

merging the ADF and LRDF (the possibility of these functions being merged is currently undergoing investigation),

- service control/data function (SCF/SDF) — performs processing and provides access to data that is specialised for particular call-related service applications.

In addition to the speculation given above, one further point of contention about the UFM within the ITU-T is the relationship between the SCF and the BC, where the BC exists without the CC (i.e. in transit exchanges). The rationale for this relationship's existence is to provide IN support, such as number translation, for the BC function. Opposition to this view is that IN support, such as number translation, should be done by the CC function in the originating serving node and that the translated number should be passed on to the BC function. For a possible way forward on this issue, see section 4.3.5.

4.3.3 Functional entity actions for B-ISDN and IN release 3

Functional entity actions are used to illustrate high-level overviews of each signalling capability contained within B-ISDN and IN release 3. These high-level overviews are expressed as a succession of information flows, showing the type and content of the flows, which describe the signalling control actions associated with each of the signalling capabilities. Essentially it is these signalling capabilities that collectively make up the set of B-ISDN and IN release 3 signalling requirements which are input into the protocol development stage to produce the signalling protocols.

4.3.4 New capabilities and features for B-ISDN aspects of release 3

A number of new call and network connection related capabilities are being described using functional entity actions for phase 1 of B-ISDN release 3. Table 5 illustrates these new capabilities. To simplify Table 5, only the capabilities that do not exist in B-ISDN releases 1 and 2 are shown. New connection types for B-ISDN release 3 are multipoint-to-point, multipoint, bi-directional point-to-multipoint, and multipoint supported by a server or bridge function. The capabilities already available for B-ISDN releases 1 and 2 have naturally been extended in release 3 to support these new connection types. It should also be noted that the capabilities associated with B-ISDN release 3 have now been extended so that they may be invoked by the IN as well as by the individual parties.

Table 5 B-ISDN release 3, phase 1 capabilities.

Signalling requirements	Connection type
Join a party to an existing call (leaf-initiated join to a call)	point-to-point, point-to-multipoint, multipoint-to-point, multipoint, bi-directional point-to-multipoint, or server supported multipoint
Join a party to an existing call and existing network connection(s) (leaf-initiated join to a call and network connection(s))	
Simultaneous call and network connection(s) transfer	
Call transfer	
Network connection(s) transfer	
Call ownership transfer	
Party ownership transfer	
Network connection ownership transfer	
Branch ownership transfer	

Additional features not relating to call and network connections are being added to phase 2 of B-ISDN release 3. These include additional capabilities such as access control to handle mobile terminals and access networks such as VB5 [17], and session control and resource control which together will provide APIs to allow enhanced multimedia applications to operate in network-rich environments. This means that applications will be able to operate over a multitude of network types such as B-ISDN, Internet or PSTN. Finally, adding standardised vertical interfaces on the B-ISDN release 3 protocol stacks will provide greater flexibility when provisioning network equipment.

4.3.5 New capabilities and features for IN aspects of capability set 3

Work on IN CS-3 is now under way and it aims to address a number of problems. The IN and B-ISDN systems have developed independently. This leads to interworking problems as the relationship between the two systems is not clearly defined and there is an overlap of services which are offered by both systems. For example, some ISDN supplementary services, such as charging and call screening, can be provided by the IN.

More recent networks have extra capabilities which the IN may need to process. Some of these are relatively simple to add, such as the extra information associated with new connection types. Other more complex areas will require further research. These include how to support multipoint/multiconnection calls, and how broadband will affect the IN call models. As there are many things that need to be added in IN CS-3 this capability set has been subdivided into CS-3.1 and CS-3.2.

IN CS-3.1 will concentrate on realistic changes that are feasible in the short time-scales planned. Therefore it will

not offer any major new features but it will build on what is already within IN CS-2 and hopefully fix problems identified such as complete IN/ISDN interworking. It should provide better support for the new network capabilities, in particular point-to-multipoint broadband connections [6]. Broadband networks will have a semi-integrated IN system in that IN will support point-to-multipoint connections, but only where the IN point of attachment is at the 'root' or point end of the point-to-multipoint connection. IN CS-3.1 will not support call and bearer separation; however, additional IN triggers and Q.2971 [6] information elements will be accommodated and supported by the SCPs. Table 6 shows the Q.2971 parameters that currently cannot be passed to the IN.

Table 6 Possible new INAP parameters.

ATM adaptation layer parameters
ATM traffic descriptor
Connection identifier
Quality of service parameter
Broadband bearer capability
Broadband low layer information (B-LLI)
Broadband high layer information (B-HLI)
End-to-end transit delay
Call state

It is expected that all these new parameters will be optional so that the network operator will be able to specify which information to export from the SSP to the SCP when triggered, if required by the IN. An example of why this data is needed would be to select a suitable SRF for the calling terminal which may now be incompatible with certain types of SRF, e.g. playing voice announcements to a set-top box should be avoided.

Other improvements that are likely in CS-3.1 include enhancements to the feature interaction manager to limit service interaction problems, and new capabilities for the SCF-to-SCF interface, particularly in the area of billing. Developments to the call-unrelated signalling should make this feature more generic and useful.

IN CS-3.2 is where all the new proposals that require longer time-scales additional to IN CS-3.1 will be considered.

The possibilities include new basic call state models (one for the call control level and one for the bearer control level) enhancing the SRF to support videoconferences, and acting as service gateways with the ability to download software to users.

Call and bearer separation

Call and bearer separation is the biggest issue facing the IN under broadband. Should the call models be split and reflect the individual state of both the bearer and the call control? At the moment the IN has the equivalent of bearer control because it needs to establish connections to SRFs and the final destination. This requirement is not likely to disappear with the migration to broadband. Similarly the equivalent of call control is occurring when the IN interacts with the caller, e.g. processing dialled digits or playing announcements.

Interaction of SCF with B-ISDN separated call and bearer state machines

IN CS-1 INAP [1] and IN CS-2 INAP [2, 18] still refer to a call as a call containing a single connection. To progress INAP into a separated call and bearer environment it is being suggested that the interaction between IN and B-ISDN is restricted to the establishment, modification and release of individual bearers associated with the party end-points. The benefit of this approach would be that the IN call model need not change and that the IN SCF would not be required to be different if the underlying protocols have or do not have the capability of controlling multiple connections per call. This issue is illustrated in Fig 6.

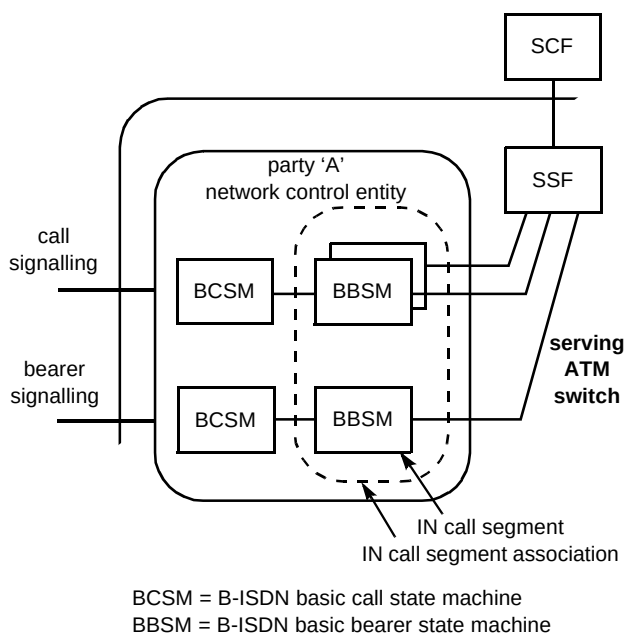


Fig 6 Separated call and bearer environment.

It is the authors' opinion that the IN CS-2 call segment is equivalent to the B-ISDN release 2/3 basic bearer state machine and that the IN CS-2 call segment association represents the set of all B-ISDN release 2/3 basic bearer state machines for a given party. Using this supposition, any enhanced INAP protocol could operate on a BBSM as if it were an IN CS-2 call segment and any IN and B-ISDN interworking may be arranged in such a way that the IN does not need to know party A's signalling protocol capabilities.

The other issue is how the basic call state models (BCSM) should be modified to support the new networks. At the moment the BCSMs are based on the telephony call model, although they should ideally be generic for all call types to enable the same application on an SCP to be used for all call types. Perhaps different call types will require unique BCSMs to allow all these new network capabilities to be utilised by the IN. The default BCSM would remain for simple services or legacy networks.

5. Conclusions

This paper has described the requirements that different users will impose on the global telecommunications network for the 21st century. It has also illustrated how some of the technology detailed elsewhere [4, 8, 12, 16, 17] will be utilised to provide these new services. Out of the many proposed alternatives for supporting these requirements just two have been examined in greater detail here.

The current intelligent network provides a good base to move towards the 'unified telecommunications network'. A wide variety of services can be built from what is already specified, and the ability for the network operator to develop their own services via a service creation system enables services currently not foreseen to be constructed relatively easily.

While a solution building on existing equipment will definitely be ready by the turn of the century it is doubtful if it will initially achieve all of the requirements placed upon it. The 'unified telecommunications network', however, is designed to handle all existing requirements and to be flexible enough to handle many future requirements. The drawback with this solution is that it is unlikely that there will be any equipment available until 2004.

So what does this mean for telecommunications users? If it means that telcos will not be able to provide public telecommunications networks to satisfy the expectations of many users in the year 2000, then the telcos have failed. However, what this most likely means is that to manage expectations a balance needs to be struck to provide users with the most pragmatic and innovative solutions based on existing equipment and emerging standards until a truly integrated telecommunications network is delivered by 2005 at the earliest. Perhaps the millennium is coming just five years too early for the 'unified telecommunications networks'.

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Gary Bruce received a BSc Honours degree in Information Technology from Leicester Polytechnic in 1988. After graduating, he joined the digital services division at BT Laboratories and has been involved in the production of signalling test equipment and the formal specification of signalling systems. In 1993, he joined the Network Intelligence Engineering Centre to develop broadband signalling systems and to standardise broadband signalling requirements within the ITU-T.

He is an Associate Member of the IEE.



Jonathan Clark received a BSc Honours degree in Electronic Engineering from Lancaster University in 1988. After graduating, he joined the digital services division at BT Laboratories and was involved in the testing of signalling test equipment and then built the Jetphone network management system communications agent. He has been involved in various IN projects and represented BT in Eurescom projects concerned with pan-European IN and harmonised IN/B-ISDN networks.

He obtained the BT MSc in Telecommunications Engineering in 1996 and is an Associate Member of the IEE.